MINOR IN MANUFACTURING AND AUTOMATION ENGINEERING

Rationale

The Manufacturing and Automation Engineering (MAE) minor aims to broaden engineering students' knowledge of up-to-date automation techniques and manufacturing methods. It also prepares undergraduate students for potential graduate studies in Manufacturing and Automation Engineering and related fields.

It primarily targets mechanical and mechatronics engineering students, but it should also accommodate students from a variety of engineering backgrounds, provided all prerequisite courses are completed.

The program mainly focuses on decision-making, materials science, manufacturing processes, quality control, system integration, automation, and many other related topics. Therefore, by the end of this minor, a student is expected to be familiar with the economic study, design, material selection, manufacturing, assembly, control, and automation of the entire engineering production process.

Program participants are required to complete at least 18 credits of related coursework. Although not required, with proper and timely advising, they may also complete their Co-op work experience or their BE project in Manufacturing and Automation Engineering, thereby enhancing their capabilities.

Program Objectives

The primary educational objectives of the MAE minor program are to:

- Give engineering students basic knowledge of the Manufacturing and Automation Engineering principles
- Equip students with the knowledge to pursue graduate studies in Industrial Engineering, Manufacturing Engineering, Automation Engineering, Materials Science, Engineering Management, and other related disciplines

Learning Outcomes

After completing the minor program, students should have the ability to:

- Apply relevant knowledge of mathematics, science, and engineering to identify, formulate, and solve practical problems related to Automation Engineering and modern Manufacturing processes
- Use modern CAD/CAM/CAE engineering tools
- Automate and control complex industrial equipment and processes
- Make apt decisions based on their comprehension of Professional and Ethical Behavior

Career Options

This minor allows MAE holders to start careers in supply chain management, product design and development, quality control and assurance, production lines, cost engineering departments, design, automation, and operation of integrated systems, and many other fields.

Program Requirements

While most relevant to mechanical and mechatronics engineering students, this minor is offered to undergraduate RHU students in the BS/BE program from different disciplines (mechanical, mechatronics, civil, electrical, computer and communications, and biomedical). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements. Proper advising is pertinent to the timely completion of the minor.

To complete the MAE minor, a student must

1) Declare an MAE minor by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the college major;
3) Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
4) Complete 18 credits of coursework as specified below

MAE Curriculum

The program offers a rich blend of fundamental sciences and practical, hands-on experience, equipping students with the essential knowledge to delve into Manufacturing and Automation Engineering. Accordingly, this minor shall include:

- Materials science and metallurgy
- Advanced/modern manufacturing processes
- Product design, optimization, and development
- Manufacturing process quality control
- Economic, environmental, and societal aspects of the industry
- Industrial equipment automation and production process control

The MAE curriculum encompasses 18 mandatory credits selected to satisfy the requirements of the proposed program objectives and learning outcomes.

Mandatory Courses

The student must complete a minimum of 3 credit hours of coursework exclusive to the MAE minor that will not count toward requirements for their major or any other minor. A description of the courses from which a student should select 18 credits is given below:

MECA 515	Modern Control Systems	3(3,0)
In this course, the student learns how to use data and correlate it with physical theory to build state-space and control models; design, build, and test controllability, feedback, observability, and stability; and perform black-box modeling in both the time and frequency domains. **Prerequisite**: MECA 440. On demand.		

MECA 541	CAD/CAM/CAE	3(3,0)
Fundamental principles of computer-aided design (CAD), computer-aided manufacturing (CAM), and computer-aided engineering (CAE); prototyping and distributed systems; specific rapid prototyping technologies; design for assembly; small batch manufacturing; continuous manufacturing; biotechnology and manufacturing. **Prerequisite:** MECH 224 and MECH 421. Annually.		

MECA 542	Industrial and Manufacturing Control	3(2,2)
This course will give an overview of programmable logic controllers (PLCs) and explores different PLC hardware components; number systems and codes; fundamentals of Logic; generation, loading and testing of PLC programs required to operate electrical motors and mechanical actuators used in industrial processes; developing fundamental PLC Wiring Diagrams and Ladder Logic Programs with an emphasis on programming timers and counters; integrated automation solutions through various programming architecture such as function-block and ladder diagrams; configuration of process visualization of a panel. **Prerequisite**: ELEC 320. On demand.		

MECA 545	Integrated Automation Lab	1(0,2)
This lab explores PID control algorithms; digital controllers and their components; tuning methods and process control; diagnosis and error search; analog value processing; communication between two or more PLCs. **Co-requisite**: MECA 542. On demand.		

MECA 586	Mechatronics Drives	3(3,0)
Switched-mode converter; steady-state voltages and currents of step-down, step-up, inverting, and other power converters; power semiconductor devices in a switching converter; discontinuous conduction mode; solve converters operating in DCM; implement transformer isolation in a dc-dc converter; design and analyze the feedback systems of switching regulators; Design Project. **Prerequisite**: ELEC 320. On demand.		

MECH 422	Manufacturing Processes	3(3,0)
General overview of manufacturing; properties of materials; dimensions and tolerances; machining; casting; sheet metal forming; injection molding; thermoforming; plastic processing; fundamentals of process performance (rate, quality, cost, flexibility); variation and quality; sustainability; overview of additive manufacturing. Project. **Prerequisite**: MECH 320. Annually.		

MECH 470	Product Design and Development	3(3,0)
Idea generation; concept generation; concept selection; functional analysis; engineering design process for systems and components; economic consideration; reliability analysis; product safety; design project. **Co-requisite**: MECH 421. On demand.		

MECH 527	Modern Material Science	3(3,0)
Advanced coverage of metals and alloys; applications and processing of ceramics; characteristics, applications, and processing of polymers; surface property enhancement; materials simulation; economic, environmental, and societal issues in materials science; biomaterials and nanoscale materials. **Prerequisite**: MECH 223. On demand.		

MECH 570	Digital Manufacturing	3(3,0)
In this course, the student will get an insight into diagnosing and correcting operational flaws; modeling and fabrication based on additive manufacturing technologies; automation technologies; production planning, process quality control; use of modern four-axis Computer Numerical Control (CNC) machines, 3D printers, and up-to-date software programs, including AutoCAD, Inventor, SolidWorks, and VisualCam. **Prerequisite**: MECH 422 and MECA 541. On demand.		

MECH 571	Optimization in Engineering Design	3(3,0)
Problem definition, cost function and constraints; local vs. global methods; deterministic vs. stochastic methods; linear vs. non-linear programming; constrained vs. unconstrained; continuous vs. discrete; gradient-based methods; combinatorial optimization techniques; applications to various engineering problems (single/multi-objective); project. **Prerequisite**: MATH 421. On demand.		